

Using Quantum Optics to Determine the Validity of Quantum Theory

Russell Dillon

April 2026

1 abstract

Einstein's explanation of the photoelectric effect caused debate over the reality of quantum mechanics. Theories emerged, which explained the behavior of light semi-classically. To address this question, we designed three experiments that characterized the nature of light. The Hanbury Brown-Twiss experiment was both replicated and modified with a BBO Crystal, and the Grangier, Roger, and Aspect (GRA) experiment was faithfully recreated. We quantified light's behavior using the second-order correlation function, $g^2(\tau)$, and compared our results with the expected values. Both HBT measurements resulted in a $g^2(0) \approx 1$. The GRA experiment yielded a $g^2(0) = 1.9 \cdot 10^{-3} \pm 3.4 \cdot 10^{-4}$, supporting the validity of quantum mechanics.

Einstein's explanation of the photoelectric effect caused debate over the reality of quantum mechanics. The semi-classical theory emerged, which treated light classically but still obtained the same results as quantum theory. Using the fact that photons cannot be divided into smaller quanta of energy, and the second-order correlation function $g^2(\tau)$, we quantified and measured the behavior of light across three experiments. We faithfully recreated the Hanbury Brown-Twiss (HBT) Experiment and the Grangier-Roger-Aspect (GRA) experiment, as well as modified the HBT experiment with a BBO crystal as a photon source. The HBT Measurement, with and without the BBO crystal, produced second-order correlation values of 1.22 ± 0.12 and 1.02 ± 0.10 respectively, while the GRA experiment produced a $g^2(0)$ value of $1.9 \cdot 10^{-3} \pm 3.4 \cdot 10^{-4}$. These results strongly support the validity of quantum mechanics and challenge semi-classical theory as the framework for subatomic physics.

2 Introduction

After Albert Einstein published his paper on the photoelectric effect in 1905, quantum mechanics became the prevailing theory for explaining the behavior of particles on a microscopic scale. Physicists Willis Lamb and Marlan Scully showed that the behavior of light could be explained by treating light classically if matter were quantized. This theory became known as the semi-classical theory and sparked much debate on the existence of photons.

Therefore, showing the existence of photons would disprove semi-classical theory. This would require the isolation of photons and show that they cannot be divided any further. This feat was achieved in the late 1900's by physicists Alain Aspect, Philippe Grangier, and Gerard Roger in their famous self-named experiment (The GRA experiment). The GRA experiment relies on generating pairs of photons that travel along different paths. One of these photons is sent directly to a single-photon detector, while the other is sent to a beam splitter, followed by another single-photon detector at the end of both possible beam paths. If photons did not exist, one would expect light to be detected at all three detectors simultaneously. The GRA experiment showed that light was rarely detected at all detectors synchronously, and semiclassical theory could not be correct.

The GRA experiment addressed two prob-

lems: generating pairs of photons and confirming that a light source is emitting single photon pairs. In order to generate photon pairs, we will use a beta-barium borate (BBO) Crystal and the process of spontaneous parametric down conversion (SPDC). SPDC is a non-linear optical process that occurs when photons enter a crystal, get absorbed, and are re-emitted as two photons whose energy sums to the energy of the incident photon. The single photon source is then confirmed via the output of a second-order correlation function, which relates the rate of simultaneous photon detection among multiple detectors.

The second-order correlation function, denoted $g^2(\tau)$, is a measure of the consistency that photons are arriving at multiple locations within a time, τ , of each other. We are interested in the case of simultaneous detection, or $\tau = 0$. The formula for $g^2(0)$ depends on the number of photon detectors. For two photon detectors, A and B, the second-order correlation function is

$$g^2(0) = \frac{R_{AB}}{R_A R_B} \quad (1)$$

Where R_{AB} denotes the rate at which photons arrive at detectors A and B at the same time, and R_A and R_B are the rates at which photons arrive at detectors A and B, respectively. For three photon detectors, T, A, and B, the second-order correlation function is

$$g^2(0) = \frac{R_{TAB} R_T}{R_{TA} R_{TB}} \quad (2)$$

And follows the same naming scheme as $g^2(0)$ for two detectors. For equations (1) and (2), values close to 0 correspond to low coincidence rates among all detectors, and therefore indicate the existence of a single photon source. Values close to 1 indicate that your photon source is emitting coherent light, and the arrival times of light are random and uncorrelated. If the value of the second-order correlation function is significantly greater than 1, then it is an indication that your light source is producing large amounts of bunched photons that arrive in groups. Values of $g^2(0)$ close to zero indicate that light behaves in accordance with quantum mechanics.

The acceptance of quantum mechanics as the primary explanation of physics has had cascading effects for modern technology. Photon detectors are used in many tools today. Some modern smoke alarms rely on detecting the emission spectra of contaminants in the air, and solar cells absorb photons to generate electricity. At the forefront of modern research is the use of photons as potential qubits in quantum computing. If light were not quantized, information could not be contained in the spatial position of a photon, and would limit the capabilities of photonic qubits.

3 Methodology

In total, we conducted three experiments: two Hanbury Brown-Twiss (HBT) variations and the GRA experiment. The HBT experiments consisted of a faithful replication of the original HBT experiment, and another where we replaced the Neutral Density (ND) filter with a BBO Crystal. Using software associated with the time tagger, a device that counts photon detections, we ran each experiment for 20 seconds. We also repeated each experiment ten times. This provided agreeable uncertainty estimation, resulting in a total collection time of 200 seconds for all three experiments. After we gathered data on photon detection counts, the count rates were calculated by dividing the total counts by the experiment time (20 seconds):

$$R_X = \frac{\text{Counts}_X}{20} \cdot \frac{\text{counts}}{\text{second}}$$

Using the counts, rates, and relations (1) and (2), the values for $g^2(0)$ were calculated for all trials, and conclusions were drawn about the theory that could explain the behavior of light observed.

4 Uncertainty

Recall that relations (1) and (2) follow the form

$$g^2(0) = \frac{R_W R_X}{R_Y R_Z}$$

We treated the second-order correlation function as

$$g^2(0) = \frac{R_{1eff}}{R_{2eff}}$$

We first calculated our uncertainty in the numerator and denominators using the calculus-based relation

$$\alpha_{R_{1eff}} = R_{1eff} \cdot \sqrt{\left[\frac{\alpha_{R_W}}{R_W}\right]^2 + \left[\frac{\alpha_{R_X}}{R_X}\right]^2}$$

and

$$\alpha_{R_{2eff}} = R_{2eff} \cdot \sqrt{\left[\frac{\alpha_{R_Y}}{R_Y}\right]^2 + \left[\frac{\alpha_{R_Z}}{R_Z}\right]^2}$$

The uncertainty in $g^2(0)$ follows from this, using the same uncertainty relation:

$$\alpha_{g^2(0)} = g^2(0) \cdot \sqrt{\left[\frac{\alpha_{R_{1eff}}}{R_{1eff}}\right]^2 + \left[\frac{\alpha_{R_{2eff}}}{R_{2eff}}\right]^2} \quad (3)$$

To determine the uncertainty in an individual count rate, R_X , we took the standard error of the mean of the corresponding count rate across our trials, defined as

$$SEM = \alpha_{R_X} = \frac{\sigma}{\sqrt{N}}$$

Where σ is the standard deviation of the counts sample, and $N = 10$ is the number of trials.

5 results

After the completion of each experiment, we obtained the following values:

Experiment	$g^2(0)$	$\alpha_{g^2(0)}$
HBT	1.02	0.10
HBT (with BBO)	1.22	0.12
GRA	$1.9 \cdot 10^{-3}$	$3.4 \cdot 10^{-4}$

Our final value of $g^2(0)$ was determined by averaging all ten trials of each experiment. The values $g^2(0) \approx 1$ for both Hanbury Brown-Twiss experiments correlate to a steady source of coherent photons, as is the case with a laser. Additionally, the GRA experiment yielded a $g^2(0)$ value close to 0, indicating the presence of single photons. To visually compare the second-order correlation function values:



Figure 1: A bar chart comparing the results of our three experiments. We observed a $g^2(0) \approx 1$ for the first two experiments, and a $g^2(0) \approx 0$ for the last

cation of single-photon pairs is consistent with the expected results from the original GRA experiment and discredits the semi-classical explanation of light. Future work could involve modifying the GRA Experiment without the BBO crystal, perhaps with light being attenuated and passed towards all three photo-detectors, similar to the HBT measurement. This result has major implications for the study of microscopic physics, where quantum mechanical effects dominate over the classical regime.

7 References

- (1) Wikimedia Foundation. (2025, September 8). Hanbury Brown and Twiss effect. Wikipedia. <https://en.wikipedia.org/wiki/HanburyBrownandTwisseffect>
- (2) Thorlabs Edu-QOP1 user manual PDF Download. ManualsLib. (n.d.). <https://www.manualslib.com/manual/3220856/Thorlabs-Edu-Qop1.html>

6 conclusion

Our measured values of $g^2(0)$ are consistent with theoretical expectations. The HBT measurements verify that the laser is a coherent light source, while the GRA experiment confirms that the BBO crystal is a reliable source of single photon pairs. Furthermore, the experimental verifi-

2

8 Supplementary Diagrams

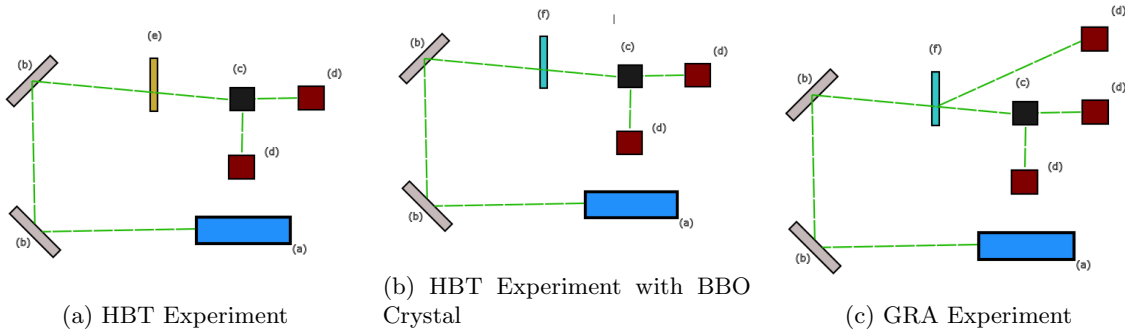


Figure 1: Experimental diagrams, from left to right, for the HBT, HBT with BBO Crystal, and GRA experiments. The pump laser (a) directs light towards two mirrors (b), towards an optical device (differing for each experiment), towards a beam splitter (c), and towards single-photon detectors (d). (e) is a neutral density filter for the HBT experiment, (f) is a BBO Crystal for the modified HBT experiment and GRA experiment. The green dashed line represents the optical path of the light emitted from the laser, and our mirrors serve to help finely align the laser with our targets.